

Albert R. Gnad

Aerospace Engineer · Computer Scientist

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*"The three most exciting sounds in the world... anchor chains, plane motors, and train whistles."
— It's a Wonderful Life*

Summary

B.S. from UW–Madison MechE, M.S. and PhD from MIT AeroAstro. Interested in sustainability, transportation (especially aviation), and the Julia programming language. Former NSF GRFP fellow. Private pilot.

Education

Massachusetts Institute of Technology

Cambridge, MA

PhD Aeronautics and Astronautics

May 2022

- GPA 5.0/5.0
- Concentration in applied & scientific machine learning (SciML)

M.S. Aeronautics and Astronautics

Feb 2018

- GPA 5.0/5.0
- Concentration in aerodynamics & air-breathing propulsion

University of Wisconsin–Madison

Madison, WI

B.S. Mechanical Engineering

May 2015

- GPA 3.98/4.0, Graduated with Highest Distinction
- Certificate in Engineering Thermal Energy Systems
- Honors in Research
- Studied abroad at the Budapest University of Technology and Economics

Work Experience

MIT CSAIL

Cambridge, MA

Postdoctoral Associate

Jun 2022 – May 2024

- Research & development of the open-source [MagNav.jl](#) software package
- Presented bleeding edge work on airborne magnetic anomaly navigation (MagNav) & aeromagnetic compensation at 6 conferences

SandboxAQ

Remote

Consultant

Jun 2022 – Sep 2022

- Advised the development & debugging of MagNav software
- Performed literature review on MagNav-related research articles

MIT AeroAstro

Cambridge, MA

Research Assistant

Sep 2015 – May 2022

- Assessed the technical & environmental viability of all-electric commercial aircraft
- Led the propulsion analysis for a short takeoff & landing (STOL) aircraft design
- Designed a test section with a flexible wall for use in a subsonic wind tunnel
- Developed ML-based compensation models for airborne magnetic anomaly navigation (MagNav)
- Collaborated with researchers across multiple labs & universities, leading to 5 publications

Wright Electric

Consultant

*Remote**Dec 2019 – Jan 2020*

- Modeled turboelectric fan performance toward development of the Wright 1 aircraft
- Evaluated the eMSTAR aircraft design & performed an analysis of alternatives

UW-Madison Eriten Research Group

Undergraduate Researcher

*Madison, MI**Sep 2014 – May 2015*

- Computed frictional energy dissipation & damping variations under various loadings
- Designed a test setup with multidimensional piezoelectric actuators & force measurements

Ford Motor Company

Refrigerant Subsystem Intern

*Dearborn, MI**May 2014 – Aug 2014*

- Reorganized & improved the functionality of a thermodynamic analysis tool
- Developed an acceptance test procedure for a new refrigerant subsystem test chamber
- Analyzed A/C performance test data to determine causes of enhanced performance
- Communicated with external teams to answer questions about refrigerant subsystems

UW-Madison Engine Research Center

Undergraduate Researcher

*Madison, MI**Jan 2014 – May 2014*

- Developed & implemented a statistical combustion model into a cycle-simulation tool
- Exercised model to explain sources of cycle-to-cycle stochastic instability in Reactivity Controlled Compression Ignition (RCCI) engines

GE Healthcare

Bearing Development Intern

*Milwaukee, WI**Jun 2013 – Aug 2013*

- Performed weekly bearing coast down data analyses for multiple programs
- Completed a tolerance stack-up analysis between a rotor & stator (included GD&T)
- Designed a bolted joint test to ensure constant bolt elongation (including solid modeling)
- Setup & monitored automated bearing tests in rotational & eccentricity rigs

UW-Madison Polymer Engineering Center

Undergraduate Researcher

*Madison, MI**May 2012 – Sep 2012*

- Designed & prototyped a cost-effective inhaler spacer made from recycled materials
- Researched ideal properties & design considerations for inhaling aerosolized medicine

GE Healthcare

Verification & Validation Co-op

*Madison, WI**May 2012 – Jan 2013*

- Oversaw automated software tests for anesthesia machines & critical care ventilators
- Created & updated (& executed) test protocols with specification changes
- Won 2 monthly awards for efficiency & best risk remediation
- Leader in reporting critical care ventilator software issues

Skills

General	Data analysis & visualization, machine learning, statistics, communication, documentation, teamwork
Programming	Julia (expert), Python (Dash/Plotly, NumPy, Pandas, PyTorch, SciPy, scikit-learn, etc.), MATLAB, Fortran
Technical	Microsoft Office , Git/GitHub , ArcGIS , TeX , Shell (Bash/Zsh), Tableau, EES, OpenVSP, SolidWorks, TASOPT, XFOIL
Private Pilot	airplane single-engine land

Honors & Awards

Achievements

2015	Passed , Fundamentals of Engineering (FE) Mechanical exam	<i>Platteville, WI</i>
2010	Valedictorian , Wisconsin Dells High School, GPA 4.0/4.0	<i>Wisconsin Dells, WI</i>
2009	Eagle Scout , designed & directed construction of a state park information kiosk	<i>Wisconsin Dells, WI</i>

Fellowships & Scholarships

2015	Dodson Fellowship , Tau Beta Pi Engineering Honor Society	<i>Madison, WI</i>
	F.M. Young Award , Pi Tau Sigma Mechanical Engineering Honor Society	<i>Madison, WI</i>
	Graduate Research Fellowship, National Science Foundation	<i>Madison, WI</i>
	Marjorie Roy Rothermel Scholarship , American Society of Mechanical Engineers Auxiliary	<i>Madison, WI</i>
	W.G. Kirchoffer Memorial Scholarship , Polygon Engineering Student Council	<i>Madison, WI</i>
2014	Engineering Student Scholarship , AfterCollege	<i>Madison, WI</i>
	Uyehara-Myers Scholarship , UW-Madison Mechanical Engineering Department	<i>Madison, WI</i>
2013	Edward F. Obert Endowment , UW-Madison Mechanical Engineering Department	<i>Madison, WI</i>
	Fred W. and Josephine H. Colbeck Scholarship , UW-Madison Polygon Eng. Student Council	<i>Madison, WI</i>
	John and Elsa Gracik Scholarship , American Society of Mechanical Engineers	<i>Madison, WI</i>
	Stabile Scholarship , Tau Beta Pi Engineering Honor Society	<i>Madison, WI</i>
2012	Alvarado Global Experience Scholarship , UW-Madison International Eng. Studies and Programs	<i>Madison, WI</i>
	David C. Spraker Scholarship , UW-Madison Mechanical Engineering Department	<i>Madison, WI</i>
	Foundation for Global Scholars Scholarship , Foundation for Global Scholars	<i>Madison, WI</i>
2011	Charles J. Marshall Scholarship , UW-Madison College of Engineering	<i>Madison, WI</i>
2010	Academic Excellence Scholarship , Wisconsin Higher Educational Aids Board	<i>Wisconsin Dells, WI</i>
	Freshman Academic Achievement Award , UW-Madison College of Engineering	<i>Madison, WI</i>
	Hold Harmless Grant , Wisconsin Higher Educational Aids Board	<i>Madison, WI</i>

Awards

2022	Transition Award , DAF-MIT Artificial Intelligence Accelerator	<i>Cambridge, MA</i>
2019	3rd Place , MIT Can Talk Oratory Competition	<i>Cambridge, MA</i>
2018	3rd Place , Siemens FutureMakers Challenge (software competition)	<i>Cambridge, MA</i>
2015	Winner , F.M. Young Award (outstanding graduating senior in UW-Madison MechE)	<i>Madison, WI</i>
	3rd Place & Best Technical Content , Old Guard Oral Presentation Competition	<i>Milwaukee, WI</i>
2014	Winner , ASME ICE Division Undergraduate Student Presentation Competition	<i>Columbus, IN</i>
	3rd Most Active Member , UW-Madison American Society of Mechanical Engineers	<i>Madison, WI</i>
2013	Most Active Member , UW-Madison American Society of Mechanical Engineers	<i>Madison, WI</i>

Conference Presentations

2024	ION Joint Navigation Conference, Magnetic Navigation Flight Testing on a C-17A by the DAF-MIT AI Accelerator JuliaCon, Real-Time Airborne Magnetic Navigation with MagNav.jl	Cincinnati, OH Eindhoven, NL
2023	IEEE/ION Position Location and Navigation Symposium, Knowledge-Informed Approaches for Airborne Magnetic Anomaly Navigation JuliaCon, Knowledge-Informed Learning in MagNav.jl for Magnetic Navigation	Monterey, CA Cambridge, MA
2022	AIAA SciTech Forum, Machine Learning-Enhanced Magnetic Calibration for Airborne Magnetic Anomaly Navigation ION Joint Navigation Conference, A Comparison of Aeromagnetic Compensation Models for Airborne Magnetic Anomaly Navigation JuliaCon, MagNav.jl: airborne Magnetic anomaly Navigation	San Diego, CA San Diego, CA Online
2021	JuliaCon, Airborne Magnetic Anomaly Navigation Enhanced with Neural Networks	Online
2019	AIAA SciTech Forum, Hybrid Turbo-Electric STOL Aircraft for Urban Air Mobility	San Diego, CA
2015	ASME Student Professional Development Conference, RCCI Cycle-Simulations with Stochastic Operating Conditions	Milwaukee, WI
2014	ASME Internal Combustion Engine Division Fall Technical Conference, RCCI Cycle-Simulations with Stochastic Operating Conditions	Columbus, IN

Leadership

Accenture Generative AI Program

Facilitator

MIT

Sep 2023 – Oct 2023

- Assisted senior-level Accenture employees in working through business challenges that incorporate generative AI

Department of Athletics, Physical Education, and Recreation

Advisory Board

MIT

Sep 2018 – May 2021

- Advised on matters of policy & procedure related to athletics, physical education, & recreation at MIT

Edgerton House

Athletics Chair

MIT

Mar 2019 – Apr 2021

- Managed the apartment building gym & initiated fitness-related activities

Hyperloop II Team

Aerodynamics Lead

MIT

Sep 2018 – Jul 2019

- Led the aerodynamic analysis & design of the shell (casing) for the MIT Hyperloop pod

American Society of Mechanical Engineers

Academic Chair, Banquet Chair, Secretary, Sophomore Representative, Design Team Member

UW-Madison

Sep 2011 – May 2015

- Heavily active member of the largest mechanical engineering club on campus

Pi Tau Sigma Mechanical Engineering Honor Society

Secretary

UW-Madison

Feb 2012 – May 2015

- Participated in professional development events

Tau Beta Pi Engineering Honor Society

Corresponding Secretary

- Volunteered at the Pi Mile Run & UW–Madison Arboretum

UW–Madison

Apr 2012 – May 2015

Energy Hub

Energy Hub Conference Committee

- Organized & participated in the Energy Hub Conference, twice

UW–Madison

Sep 2013 – Nov 2014

Division of Information Technology

Student Advisory Committee

- Discussed Division of Information Technology services, products, & initiatives

UW–Madison

Oct 2013 – May 2014

Associate Students of Madison

Office of the Registrar Student Advisory Board

- Provided input on Registrar projects

UW–Madison

Sep 2011 – Dec 2012

Extracurricular Activities

Intramurals	Basketball (captain), cornhole, dodgeball, football, hockey, soccer, ultimate, volleyball, & water polo
Rollerblading	Expert at navigating Boston’s “fun & exciting” street layout
Other	MIT GSC Hometown Presentation Initiative participant (2018) MIT & Kennedy Space Center Program participant (2017) Second Harvest Foodbank volunteer Young Scientists of America mentor

Publications

- [1] A. R. Gnad, A. B. Wollaber, and A. P. Nielsen, “Derivation and Extensions of the Tolles-Lawson Model for Aeromagnetic Compensation,” *arXiv*, pp. 1–9, 2022. [Online]. Available: <https://doi.org/10.48550/arXiv.2212.09899>
- [2] A. R. Gnad, “Advanced Aeromagnetic Compensation Models for Airborne Magnetic Anomaly Navigation,” Doctoral dissertation, Massachusetts Institute of Technology, 2022. [Online]. Available: <https://dspace.mit.edu/handle/1721.1/145137>
- [3] —, “Machine Learning-Enhanced Magnetic Calibration for Airborne Magnetic Anomaly Navigation,” in *AIAA SCITECH 2022 Forum*. San Diego, CA: AIAA, 2022, pp. 1–16. [Online]. Available: <https://doi.org/10.2514/6.2022-1760>
- [4] A. R. Gnad, J. Belarge, A. Canciani, G. Carl, L. Conger, J. Curro, A. Edelman, P. Morales, A. P. Nielsen, M. F. O’Keeffe, C. V. Rackauckas, J. Taylor, and A. B. Wollaber, “Signal Enhancement for Magnetic Navigation Challenge Problem,” *arXiv*, pp. 1–21, jul 2020. [Online]. Available: <https://doi.org/10.48550/arXiv.2007.12158>
- [5] A. R. Gnad, R. L. Speth, J. S. Sabnis, and S. R. H. Barrett, “Technical and Environmental Assessment of All-Electric 180-Passenger Commercial Aircraft,” *Progress in Aerospace Sciences*, vol. 105, pp. 1–30, feb 2019. [Online]. Available: <https://doi.org/10.1016/j.paerosci.2018.11.002>
- [6] A. W. Schäfer, S. R. H. Barrett, K. Doyme, L. M. Dray, A. R. Gnad, R. Self, A. O’Sullivan, A. P. Synodinos, and A. J. Torija, “Technological, Economic and Environmental Prospects of All-Electric Aircraft,” *Nature Energy*, pp. 1–7, 2019. [Online]. Available: <https://doi.org/10.1038/s41560-018-0294-x>
- [7] A. R. Gnad, S. Isaacs, R. Price, M. Dethy, and C. Chappelle, “Hybrid Turbo-Electric STOL Aircraft for Urban Air Mobility,” in *AIAA Scitech 2019 Forum*. San Diego, CA: MIT, 2019, pp. 1–22. [Online]. Available: <https://doi.org/10.2514/6.2019-0531>
- [8] A. R. Gnad, “Technical and Environmental Assessment of All-Electric 180-Passenger Commercial Aircraft,” Master’s thesis, Massachusetts Institute of Technology, 2018. [Online]. Available: <https://dspace.mit.edu/handle/1721.1/122501>